Responsible Software

Definition

RESPONSIBLE SOFTWARE (van de Poel, 2011)
Software **engineered in a responsible way** : about the **process**, with a focus on **human**, **active** (forward-looking) responsibility.

Definition

RESPONSIBLE ENGINEERING
Making design **decisions** responsibly : (1) goal to **do good** ; (2) **prevent avoidable negative impacts** ; (3) take into account people, other systems, social structures and the planet.

Definition

ETHICS
Principles and standards that guide human behaviour.

- **Metaethics** : what does "right" mean ?
- **Normative** : how should people act in general ?
- **Applied** : how to act ethically in a real situation ? ← **course**

Key idea

Responsibility in engineering : norms and standards (ISO, SIA), obligations (certifications), codes of conduct (ACM), **+ responsible design methodologies**.
Ethics goes beyond the law : a car meeting all regulations may still need a recall.

Explained to a goat

Responsible software = making your building decisions so that the software helps and does not hurt, even where the law says nothing.

Ethics in Technical Decisions

Example

A form with one first name, one last name, no apostrophe, names that never change : technical choices that **exclude people**.

Definition

ETHICAL ISSUE : a situation that may compromise at least one **ethical value or principle**.
ETHICAL AGENCY : capacity to decide and act in line with ethical values.

Key formulas

Ethical decision-making (Rest 1986, Schwartz 2016) :

$$\textbf{Recognize} \rightarrow \textbf{Evaluate} \rightarrow \textbf{Act}$$

Ethical sensitivity = recognize issues (awareness) + identify their impact (who, what). Grows with knowledge, experience, habits.

Lenses : desirability (customer), viability (business), feasibility (technology) **+ ethicality** (people and environment).

Explained to a goat

Even a name field can be unfair. Step 1 is to notice it ; then think ; then act.

Ethical Dilemmas

Definition

ETHICAL DILEMMA
A choice between alternatives that **all** conflict with ethical values, none overriding the others : a **wrong-wrong choice** (trolley problem).

Warning

- **Technical issue** : no value at stake.
- **Ethical issue** : one value compromised.
- **Dilemma** : values in conflict whatever you choose.

Example

- Brain-to-speech must not capture private thoughts : **ethical issue** (privacy).
- ML for medicines also finds chemical weapons : **dilemma** (**dual use**).
- Chatbot conversations sent to teachers to catch hallucinations : **dilemma** (learning safety vs. privacy / power).

Definition

DUAL USE : technologies usable for beneficial and harmful purposes. Develop (benefits, risk of harm) or not (no harm, lost benefits) ? GenAI is dual use.

EVIL heuristic (Lennerfors 2021) : **E**xit, **V**oice, **L**oyalty + **I**nsubordination.

Definition

JUDGMENT (Evaluate step) : cognitive (reason) + affective (emotion, intuition).
MORAL EMOTIONS = risk indicators. Self : guilt, shame / pride. Other : anger, contempt, disgust / gratitude, elevation.

Key formulas

Normative theories

- **Utilitarianism** : consequences, greatest good for the greatest number.
- **Deontology** : the action, universal principles ("never lie").
- **Virtue** : the person ("a courageous, trustworthy person would report it").
- **Care** : relationships, mutual responsibility.

Markkula framework : 1 identify issues, 2 get facts, 3 evaluate options with **several** theories, 4 choose and test, 5 implement and reflect.

Explained to a goat

A dilemma : every door hurts someone. Pick the least bad one with good reasons (consequences, rules, good character, relationships), and listen to your gut alarms.

Strategy : Stakeholder Analysis

Definition

STAKEHOLDER (Value Sensitive Design)
"Individuals, groups, organizations, institutions, and societies that might reasonably be affected by the technology" : positive or negative, direct or indirect, possibly **non-human**.

Key formulas

Question : who is impacted by my software ? **When** : early.

1. **Brainstorm** : daily users, geographically / socially close, activities created or replaced, maintenance, services needed to run, funding, communities of the domain.
2. **Direct** (in contact with the software) vs. **indirect** (impacted without interacting) to classify and find more.

Example

"**Paper supplier (indirect)** : the software replaces paper records, so the hospital's orders drop : affected without ever using it."

Warning

Graded : explain **how** it interacts / is impacted, **specific** to the scenario. Unclear case : state your hypothesis or argue both ways.

Explained to a goat

List everyone your app touches : those who use it, and those who never touch it but still feel it.

Strategy : Ethical Speculation

Key formulas

Escape the mirror (Black Mirror writers' room)
Part I, dark : a character hurt by the software, a pitch, a catchy title.
Part II, positive : 1-2 ethical issues, immediate and future consequences, a happy ending.

Warning

Not just "there should be a law" : what can **the designers** do ?

4 lenses : **Safety** (harm), **Fairness** (groups treated differently), **Sustainability** (planet), **Empowerment** (control given or removed).

Explained to a goat

Write the horror episode about your app, then the version where its makers fixed it.

Weekly Summary

Summary

- Responsible software = a process of responsible decisions; applied ethics; beyond law.
- Issue (one value) vs. **dilemma** (wrong-wrong); dual use; EVIL.
- Recognize → Evaluate → Act; moral emotions.
- Utilitarianism, deontology, virtue, care.
- **Stakeholder analysis** : brainstorm + direct / indirect, justified.
- **Ethical speculation** : dark story + happy ending.